

SUHAS PALLA

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EDUCATION

University of Illinois at Urbana-Champaign

B.S. in Industrial Engineering; Minor in Computer Science (in progress)

Expected May 2029

GPA: 3.77/4.00

Honors: James Scholar

Relevant Coursework: Computer Science (Java), Electronics, Differential Equations, Engineering Graphics & Design, Physics: Mechanics

TECHNICAL SKILLS

Languages: Python, Java, SQL, TypeScript, JavaScript, MATLAB, HTML/CSS

Data & ML: Data Pipelines, ETL, Data Visualization, Machine Learning, CNNs, LLMs, Model Evaluation, Pandas

Tools & Frameworks: Git, Next.js, Node.js, SQLite, Retrofit, Fusion 360 (CAD), Firebase

Concepts: Reproducible Analysis, API Integration, GD&T, Design for Manufacturability

WORK EXPERIENCE

FiPet – Software Developer

May 2026 – Present

Remote

- Built and shipped a username-search Cloud Function (TypeScript/Firebase) for friend discovery, integrating a normalized reservation data model and enforcing minimal profile output so no private user fields leak
- Wrote 11 emulator backend integration tests, covering auth, input normalization, and block-visibility edge cases; fixed test-isolation bugs affecting other suites in the process
- Resolved a cross-branch dependency between two teammates' in-progress work with the team lead before implementing, avoiding duplicated effort and merge conflicts

AI/ML Researcher – Distracted Driving Detection

Jun 2024 – Jul 2024

- Developed a computer-vision model to detect distracted-driving behaviors from image data, reasoning about a vehicle-safety problem domain
- Improved classification accuracy from 72% to 84% by tuning CNN architecture, refining preprocessing, and running iterative training cycles
- Engineered reproducible Python pipelines for dataset preparation, model training, and performance evaluation

PROJECT EXPERIENCE

Wagerly – NBA Player Performance Analytics Platform

Feb 2026 – June 2026

- Built a full-stack NBA analytics platform (Next.js, TypeScript, Node.js) on a 5-person honors team, enabling users to self-serve player performance insights
- Designed backend data pipelines that ingested 300K+ records from raw CSV into queryable structures, computing scoring trends and head-to-head history
- Integrated the Google Gemini API to generate AI-driven player insights, projected scores, and probability-based recommendations

Reverse-Engineered Drone – CAD Design & Stress Analysis (Fusion 360)

Feb 2026 – May 2026

- Reverse-engineered a commercial quadcopter as a 4-person team: disassembled the unit, measured components with calipers, and modeled assigned parts to scale in Fusion 360
- Integrated all individually-modeled parts into a single full-assembly CAD model, resolving fitment and mating constraints across team members' components
- Performed simulation-based static stress analysis (FEA) on the assembled model in Fusion 360 to evaluate structural performance under load

EXTRACURRICULAR & LEADERSHIP

ASME (American Society of Mechanical Engineers) – Design Team Member

Feb 2026 – Present

Champaign, IL

- Designing CAD housing components (Fusion 360) for pneumatic valves in an intelligent mattress-topper system, applying GD&T and design-for-manufacturability principles to produce fabrication-ready models
- Collaborating with hardware and mechanical sub-teams to align housing geometry with system-level integration requirements, structural integrity, and valve fitment

ICON (International Consulting Network) – Assistant Project Manager

Sep 2025 – Present

Champaign, IL

- Evaluated AI-driven logistics solutions, quantifying impact via an Automation/Augmentation/Speed-Gain/Error-Reduction model
- Modeled projected speed gains and error reduction across transportation workflows; presented recommendations to stakeholders